

Discrete-Event Simulator Report

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1. Design Overview

The simulator implements a peer-to-peer cryptocurrency network using a discrete-event simulation approach. The system models network topology, transaction propagation, block mining, and blockchain maintenance while considering various network conditions and peer capabilities.

2. Implementation Details

2.1 Network Topology

- The network consists of n peers, each assigned a unique ID.
- Nodes are randomly connected to between 3 and 6 other nodes, ensuring a fully connected graph. The implementation uses a ring-based approach.
- First a ring of peers is generated randomly arranging the peers. This ensures connectivity, and that all peers have developed at-least 2 random connections. After that, remaining connections are added to each peer randomly so that each peer has between 3 to 6 neighbors.
- The network is regenerated if it fails to meet connectivity or degree constraints.
- A fraction ($z0\%$) of nodes is categorized as "slow," while the remaining are "fast."
- Another fraction ($z1\%$) is classified as "low CPU" while others are "high CPU."

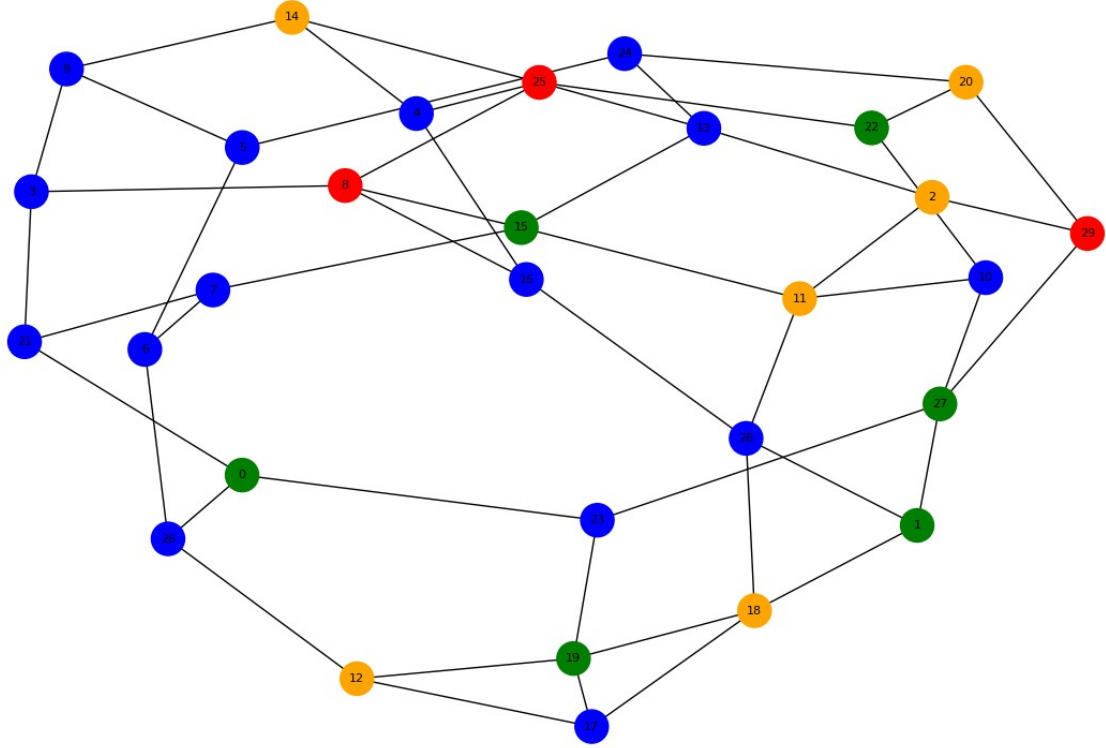


Figure 1: A sample network topology

In the above image, blue nodes are have high CPU and are on a faster connection, green are low CPU but are on fast connection, orange nodes are high CPU and are on slow connection, and red nodes are low CPU and are on slow connection.

2.2 Transaction Generation

Transactions are generated with exponentially distributed inter-arrival times. The exponential distribution was chosen for several theoretical reasons:

- It models the memoryless property of transaction generation, where the time until the next transaction is independent of how long it's been since the last transaction.
- It accurately represents independent events occurring at a constant average rate.
- It provides a good approximation of real-world transaction patterns where mining events occur identically and independently, i.e. block mining is a Poisson process, which directly translates to the time between blocks (interarrival time) being exponentially distributed

2.3 Latency Model

The network latency (L_{ij}) between peers i and j is modeled as:

$$L_{ij} = \frac{\rho_{ij} + |m|}{c_{ij} + d_{ij}}$$

where:

- p_{ij} : Propagation delay (uniformly distributed between 10ms and 500ms)
- $|m|$: Message length in bits
- c_{ij} : Link speed (100 Mbps for fast-fast connections, 5 Mbps if either peer is slow)
- d_{ij} : Queuing delay (exponentially distributed with mean $96\text{kbits}/c_{ij}$)

The queuing delay's mean is inversely related to link speed because higher capacity links can process messages faster, resulting in shorter queue lengths, whereas in case of slow links, less messages are passed per unit time, which results in less queuing on the receiver's end.

This reflects real-world behavior where congestion is more likely on slower links

2.4 Fork Resolution

If a peer receives 2 different blocks pointing to the same previous block, the peer continues mining on the block it received first.

Whenever a peer receives a block, it first constructs the chain that has this received block at the end. Then it compares the length of this candidate chain with the main blockchain he is currently mining on. If candidate chain length is greater, then he switches his main blockchain and begins mining on this new chain.

3. Experimental Analysis

3.1 Parameters Tested

- Number of peers (n): 20
- Percentage of slow nodes (z_0): 50%
- Percentage of low CPU nodes (z_1): 50%
- Mean transaction interval (T_{tx}): 5 seconds
- Mean block interval (I): 50 seconds (The mean block inter-arrival time is chosen to be just 50 seconds instead of the 600 seconds as bitcoin so as to increase the mining rate, and hence higher chance for forks, for better analysis).
- Maximum block size: 1 MB
- Simulation time: 600 seconds

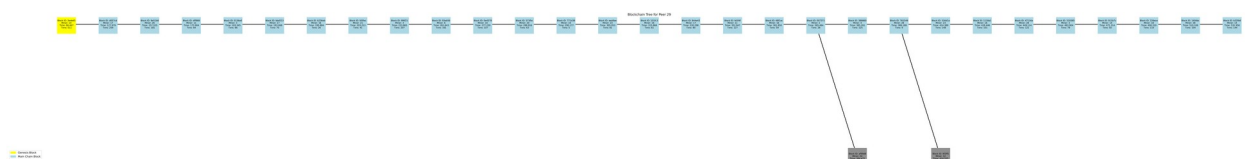


Figure 2: A typical blockchain tree

3.2 Block Generation Analysis

The ratio of blocks in the longest chain to total blocks generated varies significantly based on peer characteristics:

1. High CPU, Fast Network Peers:

- Highest success rate in getting blocks into the longest chain
- Better block propagation due to fast network connections
- Higher hash power leads to more frequent block mining

2. High CPU, Slow Network Peers:

- Moderate success rate
- Network latency causes some blocks to arrive late and become stale
- Mining power advantage partially offset by propagation delays

3. Low CPU, Fast Network Peers:

- Lower success rate due to reduced mining power
- Good block propagation helps maintain chain consistency
- Fewer blocks generated overall

4. Low CPU, Slow Network Peers:

- Lowest success rate
- Combination of slow mining and high latency
- Most likely to have blocks become stale

3.3 Tree Structure Analysis

Branch lengths in the blockchain tree show interesting patterns:

1. Main Chain:

- Consistently grows throughout the simulation
- Shows steady block addition rate
- Dominated by high CPU peers' blocks

2. Fork Branches:

- Typically 1-3 blocks long

- More common during periods of high network latency
- Often initiated by slow network peers
- Usually resolved within 2-3 block intervals

4. Key Observations

1. Network Impact:

- Network speed has a significant impact on fork frequency
- Slow network peers create more temporary forks
- Fast network connections help maintain blockchain consistency

2. Mining Power Distribution:

- High CPU peers dominate the longest chain
- Low CPU peers contribute more to temporary forks
- The 10:1 hash power ratio creates clear mining advantages

3. Block Propagation:

- Fast network peers help stabilize the network
- Slow peers can cause temporary network partitions
- Block size affects propagation time significantly

4. Fork Resolution:

- Most forks resolve within 2-3 blocks
- Longer forks are rare but possible
- Fork resolution favors fast network peers

5. Conclusions

The simulation demonstrates several key aspects of P2P cryptocurrency networks:

1. The importance of network quality in maintaining blockchain consistency
2. The significant advantage of higher hash power in block mining success
3. The relationship between network parameters and fork frequency
4. The self-stabilizing nature of the longest chain rule

The results suggest that both network capability and computational power play crucial roles in the efficiency and security of the cryptocurrency network, with network capability being particularly important for maintaining consistency across the network.

A table supporting our results for a test run is shown below.

Peer ID	LC Length	Blocks in LC by Peer	Total Blocks by Peer	Ratio (Blocks in LC/Total Blocks)
0	29	0	0	0.00000
1	29	0	0	0.00000
2	29	1	1	1.00000
3	29	3	3	1.00000
4	29	1	1	1.00000
5	29	0	0	0.00000
6	29	1	1	1.00000
7	29	0	0	0.00000
8	29	0	0	0.00000
9	29	0	0	0.00000
10	29	2	2	1.00000
11	29	2	2	1.00000
12	29	0	1	0.00000
13	29	0	0	0.00000
14	29	3	3	1.00000
15	29	0	0	0.00000
16	29	4	4	1.00000
17	29	3	3	1.00000
18	29	1	1	1.00000
19	29	0	0	0.00000
20	29	1	1	1.00000
21	29	1	1	1.00000
22	29	0	0	0.00000
23	29	2	2	1.00000
24	29	0	1	0.00000
25	29	0	0	0.00000
26	29	1	1	1.00000
27	29	0	0	0.00000
28	29	3	3	1.00000
29	29	0	0	0.00000

Figure 3